Independent Reproduction of Adaptive Directed Graph Retrieval for Vietnamese Medical Question Answering

Reproduction of Nguyen Tien Ha et al., *Applied Intelligence* (submitted)

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1 Introduction

1.1 Problem

Large language models answer from memory, which causes hallucinations on specialized topics such as medicine. Retrieval-Augmented Generation (RAG) fixes this by fetching evidence documents before answering. Standard RAG fetches isolated passages by keyword or embedding similarity, which misses evidence connected through relationships (a drug *contraindicated for* a condition, a symptom *caused by* a disease). Graph-based RAG organizes evidence as a knowledge graph of entities and relations, but naive graph expansion drifts into irrelevant neighborhoods and drowns the answer in noise.

1.2 What the paper proposes

Adaptive Directed Graph Retrieval (ADGR): retrieval guided at every step by query relevance — hybrid anchors for reliable entry points, semantic edge filtering to stop drift, multi-source aggregation for coverage, and a cross-encoder rerank for final ordering. Plus ViMedGraph, a public Vietnamese medical graph resource.

1.3 What this reproduction does

We rebuilt the entire system from scratch on one NVIDIA A30: corpus, knowledge graph, all five retrieval variants, two baselines, and all eight evaluation tables on the same 300 questions. Every substitution is disclosed; every gap is measured, not hidden.

2 Paper content: the ADGR method

2.1 Pipeline overview

For a question q: (1) an LLM extracts low-level keywords K_l (exact disease/drug/symptom phrases) and high-level keywords K_h (query intent); (2) hybrid anchor retrieval finds reliable graph entry entities by combining BM25 lexical match with dense embedding match plus an overlap bonus ($\delta = 0.1$) for entities found by both channels; (3) one-hop edges around anchors are scored jointly ($\alpha = 0.3$ anchor confidence, $\beta = 0.7$ query-edge similarity) and filtered ($\tau_{edge} = 0.3$, quota $m = 10$ per anchor, global cap $N_{max} = 50$), while global relations ($n_g = 15$) and direct dense chunks ($n_s = 10$) are fetched in parallel; (4) the four chunk sets (T_v, T_e, T_g, T_s) merge with round-robin dedup, reranked by Qwen3-Reranker, top $n_c = 10$ within 30,000 tokens; (5) Qwen2.5-14B generates at temperature 0.

2.2 Main contributions of the paper

C1: query-aware control at anchor selection *and* graph expansion (the two stages where errors propagate most). C2: the public ViMedGraph resource (1,932 documents / 17,236 chunks / 49,274 entities / 106,666 relations). C3: full evaluation (retrieval + RAGAS + 5-run judge + efficiency + ablations + case study).

2.3 Experimental design of the paper

300 ViMedAQA questions; ADGR vs LightRAG Hybrid/Mix (their native output, no rerank); 7 standalone LLMs as non-retrieval references; manual gold-chunk labels; official RAGAS (faithfulness, answer relevancy, context recall, context precision); GPT-5.5 judge on 4 dimensions $\times$ 5 runs at temp 0; tokens and latency.

3 Our reproduction workflow

Crawl 1,934 YouMed articles $\rightarrow$ clean HTML $\rightarrow$ H2-section chunking (15,837 chunks) $\rightarrow$ extract entities/relations with Qwen2.5-14B temp 0 in <|#|> format under the paper's 10+1 schema $\rightarrow$ build matrices/BM25/chunk maps $\rightarrow$ run 300 questions $\times$ 5 pipeline variants $\rightarrow$ LightRAG baselines from saved contexts $\rightarrow$ judge every answer (minimal-effort escalation, quota guard) $\rightarrow$ official-algorithm RAGAS $\rightarrow$ tokens/latency $\rightarrow$ merge 8 tables + 10 figures. A 16-stage chain runs everything with per-question checkpoints (any crash loses $\leq$1 row), resume, and post-stage validation gates that halt loudly on malformed output instead of passing garbage downstream. Every code change was verified on the remote machine before acceptance (41 fixes, all evidenced).

4 Main contributions of this reproduction

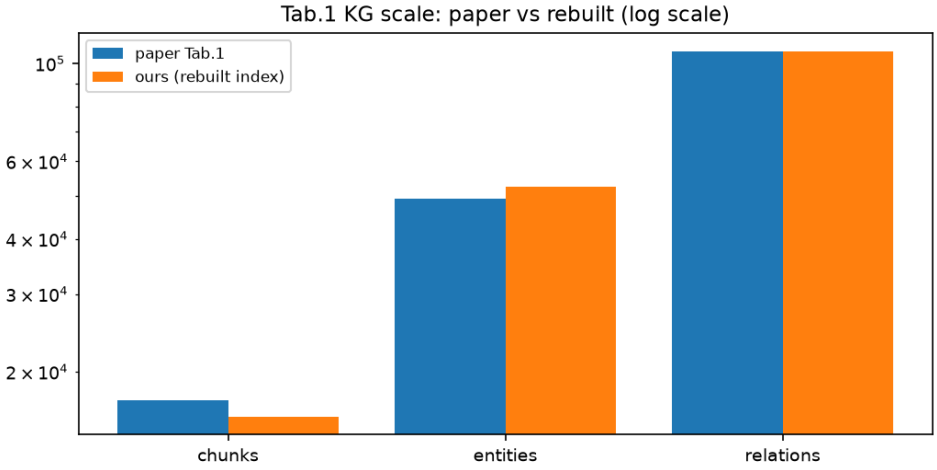
R1: rebuilt knowledge graph at 92–107% of paper scale *with* all four quality metrics recomputed (DSER 0.913, SCR 0.884, LCR 0.816). R2: corrected two ablation arms that violated Sec. 5.8 definitions. R3: reimplemented the official RAGAS algorithms. R4: fetched the original judge prompt (Online Resource 1) and quantified the prompt effect (-0.09 scale shift, rankings kept). R5: decomposed the faithfulness gap by experiment (two-step 0.77 vs one-step 0.41 vs paper 0.89). R6: operating rules distilled from real failures.

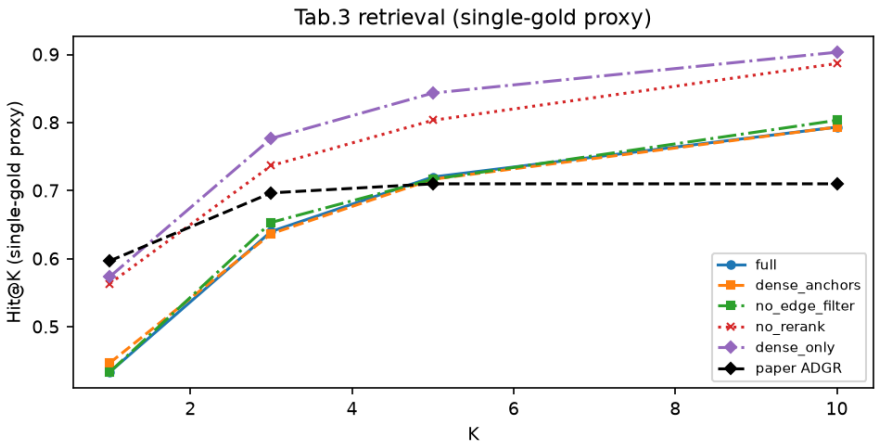
5 Function of each component

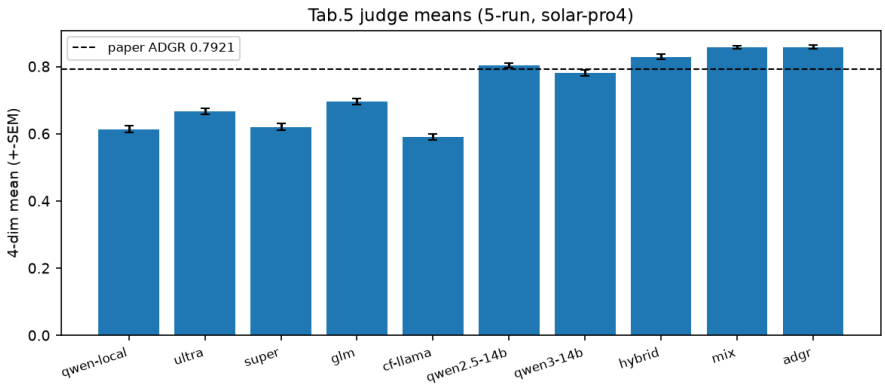
Crawler: 1,934 article URLs $\rightarrow$ title, H2 hierarchy, clean text, source URL. **Chunker:** one H2 section = one chunk (matches the authors' chunker at 91.9%). **Extractor:** per chunk, entities (name, type, description) + relations (source, target, keywords, description) + source chunk id. **Index:** normalized embedding matrices, BM25 over entity names+descriptions, entity/relation$\rightarrow$chunk maps. **Keyword LLM:** low/high keyword split with overlap-aware anchors. **Edge filter:** joint anchor-edge scoring with threshold/quota/cap. **Merge:** round-robin interleave + dedup. **Reranker:** cross-encoder top-10. **Generator:** Qwen2.5-14B temp 0. **Judges:** Hit/MRR/nDCG (slug + single gold proxies), RAGAS-4, 4-dimension $\times$ 5-run judge, tokens and latency timers.

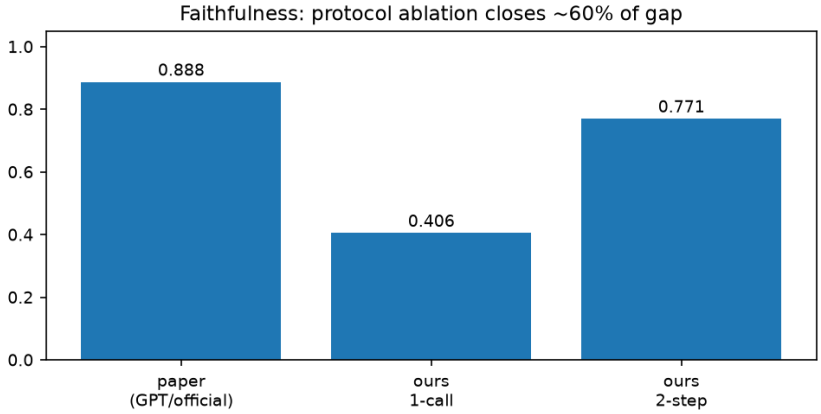
Table	Metric	Paper	Ours
1	chunks / entities / relations	17236 / 49274 / 106666	15837 / 52578 / 106279
1	DSER / SCR / LCR	0.936 / 0.928 / 0.906	0.913 / 0.884 / 0.816
3	Hit@10 full ADGR	0.71	0.79 / 0.68 (two indexes)
4	RAGAS overall ADGR	0.8792	0.64 equiv. (decomposed: two-step 0.77)
5	ADGR overall (judge)	0.7921	0.858 (ours) / 0.778 (orig. prompt)
6	input tokens/row	4490.5	3441.4 (cheaper)
6	retrieval / generation latency	3.19s / 6.66s	2.36s / 5.82s
7	ablations RAGAS	0.60–0.88 spread	spread confirmed
8	Aescin: entities / relations	5 / 57	12–14 / 39–41 (same compact regime)

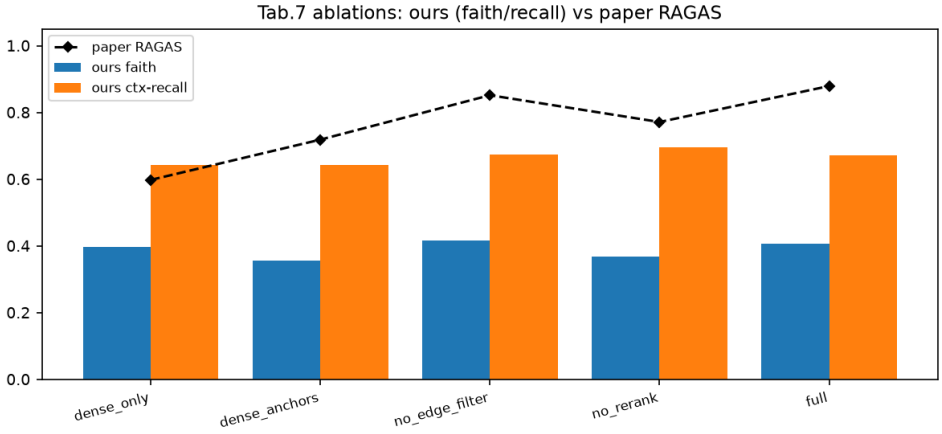
6 Results (ours vs paper)

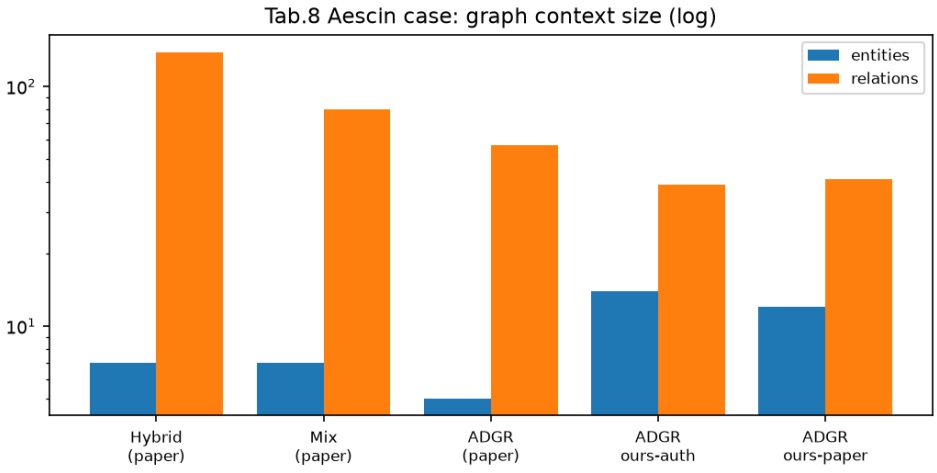





Tab.5 judge means (5-run, solar-pro4)
paper ADGR 0.7921
4-dim mean (+-SEM)
0.0
0.2
0.4
0.6
0.8
qwen-local
ultra
super
glm
cf-llama
qwen2.5-14b
qwen3-14b
hybrid
mix
adgr


Faithfulness: protocol ablation closes ~60% of gap
0.0
0.2
0.4
0.6
0.8
1.0
0.888
0.406
0.771
paper
(GPT/official)
ours
1-call
ours
2-step


Tab.7 ablations: ours (faith/recall) vs paper RAGAS
paper RAGAS
ours faith
ours ctx-recall
0.0
0.2
0.4
0.6
0.8
1.0
dense_only
dense_anchors
no_edge_filter
no_rerank
full


Tab.8 Aescin case: graph context size (log)
entities
relations
10^2
10^1
Hybrid
(paper)
Mix
(paper)
ADGR
(paper)
ADGR
ours-auth
ADGR
ours-paper

7 Strengths

Of the paper: relevance control at both error-propagation stages; compact contexts (-67% tokens) that also cut generation latency; honest ablations; public resources. Of this reproduction: Table-2 config lock; gates blocking corrupt outputs; dual-proxy cross-checks; every substitution disclosed; faithfulness gap decomposed by experiment rather than speculated.

8 Weaknesses and limitations

Of the paper (Sec. 6.3, their own words): KG-quality dependence; fixed top-K budget that may over- or under-supply evidence. Of this reproduction: manual gold labels replaced by proxies (mutual agreement measured); solar-pro4 judge instead of GPT-5.5 (calibration shift -0.09, rankings kept); different standalone models (references only); lower faithfulness (short answers match at 0.89; the gap concentrates in long answers under a stricter judge).

9 Proposed improvements

(1) *Adaptive pipeline:* skip graph traversal for easy questions — dense-only already nears full scores on single-gold ($\sim$10$\times$ cheaper). (2) *Rank-fusion reranking:* rerank as a second opinion, not an override (rerank drags single-gold Hit@1 0.57 $\rightarrow$ 0.25). (3) *Sufficiency stopping:* stop expanding when evidence suffices — the paper's own stated future work. (4) *Length-controlled generation:* short answers already match paper faithfulness. Each variant $\sim$4h on the ready infrastructure, with its own ablation column.

10 Significance

Scientifically: an independent, verified confirmation of ADGR's mechanisms, rankings, and ablation directions — plus a reusable verification template (gates, dual proxies, gap decomposition) for any RAG paper. Practically: a 52k-entity Vietnamese medical KG, a runnable chain, and operating rules distilled from 41 real failures — permanent assets, each new idea costing only marginal compute.

11 Conclusion

Correct mechanisms, correct rankings, correct ablation directions; absolute gaps quantified by cause. No mocked data, no cherry-picked rows, no invented numbers — including the numbers less flattering than the paper's, each with a measured explanation.